



SWARAJ INDIA GROUP

TECHNICAL DATA SHEET

DESIGN OF REINFORCEMENT REBAR OUPLER

{FORGED PARALLEL THREADED Class - H}

REBAR SIZE	OD (D) SIZE(±1 mm)	LENGTH (L) (±0.2mm)	FINISH WEIGHT	THREAD PITCH(mm)	THREAD SIZE
16MM	26.0	36	0.08	3.0MM	M-20X3.0
20MM	32.0	42	0.15	3.0MM	M-24X3.0
25MM	40.0	52	0.30	3.0MM	M-30X3.0
32MM	50.0	65	0.60	3.0MM	M-36X3.0
40MM	63.0	90	1.70	4.0MM	M-44x4.0

REBAR TECHNICAL SPECIFICATION..

!QUALITY PROCESS - FINAL INSPECTION							
CHECKING PARAMeTERS	jACCEPTABLELIMIT						
	16MM	20MM	25MM	32MM	40MM	CHECKING INSTRUM.ENT	FREQUENCY CHECK
OD SIZE	26.0 (±1MM	32.0(±1mm	40.0(±1mm	50.0(±1mm)	63.0(±1mm)	VERNIER CALLIPER	Pert-ts 2
LENGTH	36 (±0.smm)	42 (±0.smm)	52 (±0.smm)	65 (±0.smm)	90(±0.smm)	VERNIER CALLIPER	Pert-ts 2
PITCH(MM)	3.0mm	3.0mm	3.0mm	3.0mm	4.0mm	PLUGGUAGE	PerHrs2
THREAD	M-20X2.5	M-24X3.0	M-30X3.0	M-36X3.0	M-44X4.0	THREAD PLUG GUAGE	5Pcs Per Sh
ANTIRUST	APPLY	APPLY	APPLY	APPLY	APPLY	VISUAL	EACH
PACKING	300	200	80	50	30	VISUAL	EACH
U.T.S(N/mm²)Minimum	715	715	715	715	715	3PERLOT	
Result	TMT Bar Should Break Out Side of splice leng.						

Add:- sonawane wasti, chikhali, Pune 41, Pune 41 Mo:- 8605379932

High Tensile Strength: FeSSO / FeSSOD

As **SWARAJ INDIA GROUP** is in the market parallel threaded mechanical splicing system designed for the connection of concrete reinforcing bars of 16mm to 40 mm manufactured adopting international quality conforming to Indian & American standard **IS 1786:2008, CED 54(7589) & IS 16172:2023** by manufacturing in 2 steps like cutting, & Threading having wide spread benefits easy installation with no torque wrenching required & with numerical control (CNC) for making couplers.

The mechanical splice connection develops in tension or compression more than 100% of the specified yield strength of the concrete reinforcement bar. The connection designed under strict quality assurance by the ACP to withstand minimum breaking strength of 715 N/MM². Thus the Mechanical splicing systems is the most reliable, economical & having additional advantages & have been designed friendly with assurance of maintaining load path in continuity of the structural reinforcement, independent of the conditions or the existence of the

Cold Forging Upsetting Machine



Rebar Threading BST Machine

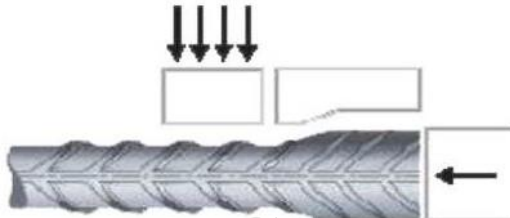


Process



Cutting:

- 1) The end of the TMT bars is must be saw Cut 90 Degree.



DC: ____.
tttt

Cold Forging:

- 2) With the help of patented cold forging, the sawn end of the reinforcing bar is then enlarged. With this the core diameter of the bar is increased to pre detennined diameter.



Threading:

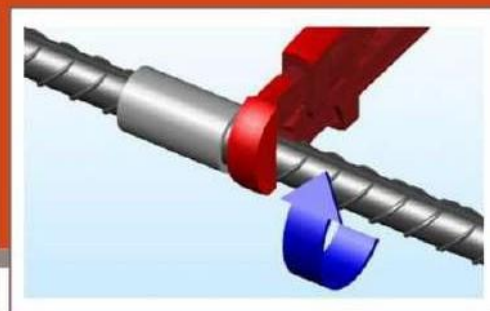
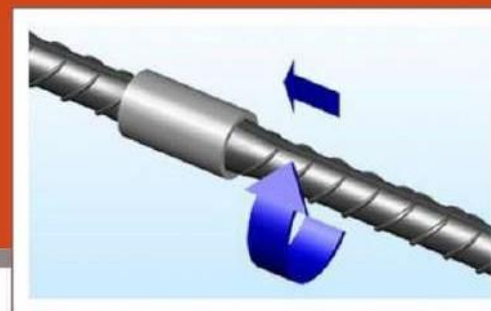
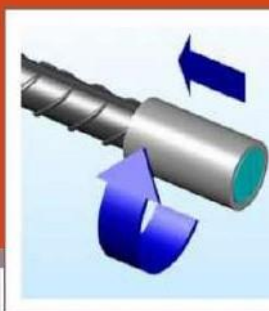
- 3) An ISO standard metiic thread is then cut.

Technical Specifications (All sizes in MM)

Size	Thread	Diameter (mm)	Forging Le, (mm)	MM Pitch	No Of Thread
16mm	M20x3	19.0 min.	18 + hnm	3.0	6-7
20mm	M24x3	23.0 min.	21 + hnm	3.0	7-8
25mm	M30x3	28.0 min.	26+ hnm	3.0	8 - 9
32mm	M36x3	35.0 min.	32.5 + 1mm	3.0	10 - 11
40mm	M44x4	44.0 min.	45 + hnm	4.0	11 - 12

i- R e h a r C o u p l e r I n s t a l l a t i o n

Type 'B' System Parallel Thread



1] Screw the coupler to the rear of the thread on the fixed bar and lock tight. The bar end should be central within the coupler.

2] Next Bar put and rotate the continuation bar in the coupler.

3] Tighten the joint using a wrench on the continuation bar. After tightening there should be no more than 2-4mm of thread exposed, depending on the diameter of the rebar.

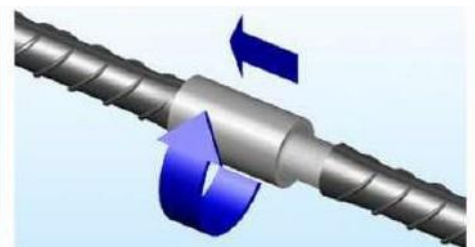
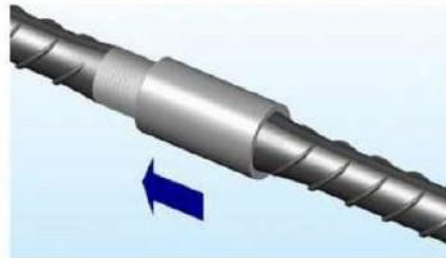
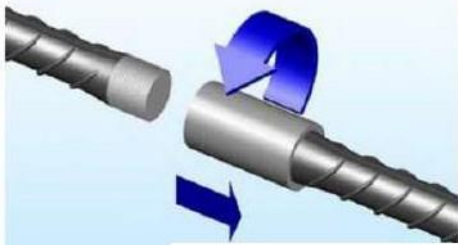
1] कपलरला निश्चित बारवरील धाग्याच्या मागील बाजूस स्कू करा आणि घट्ट लॉक करा. बारचा शेवट कपलरच्या मध्यभागी असावा.

2] नंतर बार घेऊन कपलरमध्ये घुंटी घ्या.

3] जोड जोडण्यासाठी कपलरवरील बारवर वॉल्व्हन वापरा. जोड जोडल्यानंतर 2-4 मिमी थ्रेड बाहेर राहणे आवश्यक आहे, जोड जोडण्यासाठी बारचे व्यास.

i_R_e_h_a_r_C_o_u_p_l_e_r_I_n_s_t_a_l_l_a_t_i_o_n

Type 'B' System Parallel Thread



1] Screw the coupler to the rear of the thread on the continuation bar.

2] Position the continuation bar with the coupler against the end of the first bar.

3] Rotate the coupler from the continuation bar to engage against the rear of the thread on the opposing bar and lock tight.

1] ci.ilN.fta qa 'oDcyaa
qiOci-.l cITTL.

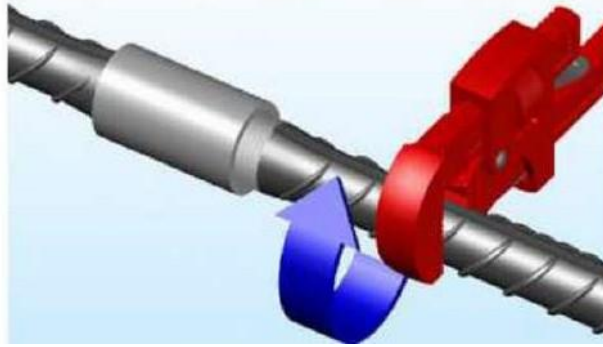
2] पहिल्या बारच्या समाप्तीच्या विरुद्ध कपलरसह फिरवा.

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ct5g a?ai cITTR?H?tff
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Rebar Coupler Installation



Type 'B' System Parallel Thread



4] Using a wrench, rotate the continuation bar to lock the two bar ends against each other within the coupler. After tightening, the length of exposed thread should be no more than half of the coupler length plus 2-4mm depending on the diameter of the rebar. IT IS NOT NECESSARY TO USE A TORQUE WRENCH DURING ASSEMBLY

4] पाना वापरून, जोड्यामध्ये दोन्ही बार एकमेकांच्या विरुद्ध लॉक करण्यासाठी सुरू पट्टी फिरवा. कडक केल्या नंतर, रिबारच्या व्यासावर अवलंबून उघडलेल्या धाग्याची लांबी कपलरच्या लांबीच्या अर्ध्यापेक्षा अधिक असू नये. असेन्सीच्या दरम्यान एखादी टाक्यू रच वापरणे आवश्यक नाही.

Rebar Coupler Installation



Type 'B' System Parallel Thread



No Gap between two thread

5] Inspect the splice for proper thread engagement. For Bar splicer threads, some variation in the number of exposed threads is natural due to thread tolerance build-up and thread run-out. In general, it is usual to see 0 to 1 threads on the position setting bar after full assembly

(Note: Position splice bar should have approximately half of its overall thread length visible after proper assembly). **IT IS NOT**

NECESSARY TO USE A TORQUE WRENCH DURING ASSEMBLY OR APPLY A HIGH TORQUE VALUE

5] थ्रेडच्या योग्य प्रतिबद्धतेसाठी स्प्लिसची तपासणी करा. बार्सप्लिकर थ्रेड्ससाठी, उघडलेल्या थ्रेड्सच्या संख्येमध्ये काही भिन्नता नैसर्गिक आहे थ्रेड टॉलरेंस बिल्ड-अप आणि थ्रेड रन-आउट. सर्वसाधारणपणे, पूर्ण असेंब्लीनंतर स्थिती सेटिंग बारवर 0 ते 1 थ्रेड दिसणे नेहमीचेच आहे (टीप: स्थितीत स्प्लिस बारमध्ये त्याच्या संपूर्ण धाग्याच्या लांबीच्या अर्ध्या भागाच्या योग्य असेंब्लीनंतर दृश्यमान असावी). असेन्सीच्या दरम्यान एखादी टाक्यू रच वापरणे आवश्यक नाही किंवा उच्च टाक्यू व्हॅल्यू लागू नाही